A

Minor Project Report on

POLLU-SENSE

AI-Based Air Quality Prediction and Monitoring System

At

Poornima College of Engineering, Jaipur

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Information Technology

(Session 2026-2027)

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October, 2026

CANDIDATE DECLARATION

We hereby declare that the work which is being presented in this Minor Project Report entitled **"POLLU-SENSE: AI-Based Air Quality Prediction and Monitoring System"**, in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology in Information Technology**, and submitted to the Department of Information Technology, **Poornima College of Engineering, Jaipur**, is an authentic record of our own work carried out under the guidance of Mr. Arpit Jain and Mr. Alok Singh during the session **2026–27 (Odd Semester)**.

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

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Place: Jaipur
Date: 25/10/2026

Enclosed: Project Presentation (Pollu-Sense) and Demonstration Video Link

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Date: 25/10/2026

DEPARTMENT CERTIFICATE

This is to certify that **Minor Project** report titled **POLLU-SENSE: AI-Based Air Quality Prediction and Monitoring System** has been submitted by **Nirmal Todwal (PCE23IT035), Avisha Goyal (PCE23IT011)**, **Sahil Rochwani (PCE23IT050)** and **Aditi Jain (PCE23IT004)** in partial fulfillment for the award of the Degree of **Bachelor of Technology** in **Information Technology** during the session 2026-27, Odd Semester. The minor project work is found satisfactory and approved for submission.

Names & Signatures

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Date: 25/10/2026
Place: Jaipur

ACKNOWLEDGEMENT

We are sincerely grateful to **Mr. Arpit Jain** and **Mr. Alok Singh**, Faculty Coordinators of the Minor Project, Department of Information Technology, Poornima College of Engineering, for their invaluable guidance, regular feedback and encouragement throughout the development of this project. Their mentorship has been instrumental in shaping the innovative aspects of this initiative.

We wish to express our profound gratitude to **Dr. Gajendra Singh Rajawat**, Head of the Information Technology Department at Poornima College of Engineering, for his unwavering encouragement, guidance, and motivation during the course of this work.

A special thanks to **Dr. Mahesh Bundele**, Director and Principal of Poornima College of Engineering, for fostering a conducive learning environment and providing the necessary resources for this project.

We also wish to acknowledge the visionary leadership of **Ar. Rahul Singhi**, Director of Poornima Group, whose support has significantly enriched our academic journey and facilitated the successful completion of this project.

This project, **POLLU-SENSE**, is a testament to the collective efforts of all involved, and we sincerely thank everyone for their contributions. We are also thankful to our families and friends for their constant support and motivation.

Thank You

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ABSTRACT

This project presents POLLU-SENSE, an AI-based air quality prediction and monitoring system that tracks pollution in real time, predicts future pollution spikes, identifies the probable source of pollution and automatically notifies the responsible parties and authorities. The platform is built with modern web and data technologies: React.js and Tailwind CSS for an interactive and responsive front end; Django and Flask (Python) for the back-end services and real-time APIs; TensorFlow and Scikit-learn for machine-learning based prediction; and PostgreSQL and Firebase for data storage, authentication and notifications.

The system collects air quality data from IoT sensors, the OpenWeatherMap API and Central Pollution Control Board (CPCB) data sources and displays it on a live dashboard and an interactive map. When pollution crosses a configured threshold, an automated alert is generated and sent by e-mail to the concerned industry or individual. If no corrective action follows, the case is escalated to the regulatory authorities. A newsletter subscription service, a crowdsourced reporting facility and an industry compliance scorecard extend the platform from simple monitoring to active pollution management.

The project demonstrates how artificial intelligence, the Internet of Things and cloud services can be combined into a practical, scalable and low-cost prototype that supports smart-city environmental governance. It reflects the application of full stack development, machine learning, RESTful API design and asynchronous data handling to a socially relevant problem, and can be extended to multiple cities and different pollution sources.

Key features:

- Real-time air quality monitoring with a live dashboard and interactive pollution map.
- AI-based prediction of future pollution levels and spikes.
- Source identification: traffic, industry, construction or natural causes.
- Automated e-mail alerts with an escalation mechanism for authorities.
- Crowdsourced reporting, newsletter subscription and health protection advice.
- Industry compliance scorecard with historical pollution tracking.
- IoT-ready design for smart purification measures such as mist sprays and filters.

Chapter 1

Introduction

POLLU-SENSE is an AI-based platform developed to monitor air quality continuously, predict pollution spikes before they occur, identify the likely source of pollution and alert the people responsible for taking corrective action. Air pollution is one of the most serious environmental and public-health challenges faced by Indian cities today. Yet the present monitoring arrangement is largely periodic, sparse and reactive, rather than continuous, dense and preventive.

The primary objective of this project is to design and implement an intelligent, automated and user-friendly system that connects sensors, public air-quality data, machine-learning models and citizens on a single web platform. Pollu-Sense converts raw readings of pollutants such as PM2.5, PM10, NO_2, SO_2, CO and O_3 into an easily understandable Air Quality Index (AQI), shows them on a live dashboard and map, forecasts how they are likely to change and triggers alerts when safe limits are crossed.

Traditionally, pollution control boards have relied on periodic reports and a limited number of fixed monitoring stations. High pollution levels are detected but are rarely attributed to a specific polluter, and the absence of instant alerts means that industries and authorities often react late. The result is long-term damage to the environment and to public health. Pollu-Sense overcomes these limitations by providing real-time monitoring, AI-driven source identification, automated notifications and a transparent escalation path to the regulating authorities.

The project follows a full stack development approach. React.js and Tailwind CSS are used for the front end; Django and Flask provide the back-end services and the real-time alert API; TensorFlow and Scikit-learn are used for the prediction models; and PostgreSQL, Firebase, the OpenWeatherMap API, CPCB data and the Google Maps API supply storage, authentication, data and visualization.

1.1 Purpose of the System

The main purpose of developing Pollu-Sense is to create an intelligent, transparent and scalable digital platform that bridges the gap between environmental data and timely action. It aims to turn air quality monitoring from a passive, report-based activity into an active, alert-driven process that involves industries, citizens and regulators together.

For citizens, the platform provides easy access to real-time air quality, health protection guidance and a channel for reporting pollution sources. For industries and local bodies, it provides early warnings and a compliance scorecard. For authorities, it provides a centralized, evidence-based view of pollution hotspots and repeat violators, which reduces manual effort and speeds up regulatory intervention.

1.2 Objectives of the Project

The primary objectives of the Pollu-Sense system are:

1. To design and develop a web-based air quality monitoring platform that displays real-time pollution data on a dashboard and an interactive map.
2. To collect data from IoT sensors, the OpenWeatherMap API and CPCB sources, and to store it for historical analysis.
3. To apply machine-learning models to predict future pollution levels and detect abnormal spikes in advance.
4. To identify whether pollution originates from traffic, industry, construction or natural causes.
5. To send automated alerts to the responsible industry or individual and to escalate unresolved cases to the authorities.
6. To encourage public participation through crowdsourced reporting, a newsletter service and health protection information.

1.3 Problem Statement

Problem Title: AI-powered system for real-time air pollution tracking, notification and control.

Problem Description: The project aims to develop an AI-integrated real-time Air Pollution Tracking System that keeps track of potential air pollution-causing agents. The AI-powered system continuously tracks city-wide pollution, identifies spikes, and triggers instant alerts. By integrating IoT and predictive analytics, it ensures proactive measures for cleaner air so that future generations get a fresh and clean air to breathe in.

1.4 Project Details

Project Title	POLLU-SENSE: AI-Based Air Quality Prediction and Monitoring System
Category	Minor Project (Odd Semester)
Department	Department of Information Technology
Institute	Poornima College of Engineering, Jaipur
University	Rajasthan Technical University, Kota
Session	2026-27
Faculty Coordinators	Mr. Arpit Jain, Mr. Alok Singh
Team Members	Nirmal Todwal, Avisha Goyal, Sahil Rochwani, Aditi Jain

1.5 Scope of the Project

- Real-time monitoring of AQI, temperature and humidity for multiple locations on a map and dashboard.
- Prediction of near-future pollution trends from historical and live data.
- Alerts by e-mail, a newsletter subscription service and a contact channel for the public.
- A modular design that can start in highly polluted cities and later expand to more cities, sensors and pollution sources.

1.6 Need for the System

Air pollution is a major cause of respiratory infections, cardiovascular disease and lung cancer, and it affects children, the elderly and people with existing conditions the most. The World Health Organization estimates that air pollution is linked to about 7 million premature deaths every year worldwide [4]. India bears a very large share of this burden and also suffers heavy economic losses through healthcare costs and lost labour productivity [5]. Table 1.1 summarizes the figures that motivated this project.

Indicator	Reported value	Source
Premature deaths linked to air pollution (global)	About 7 million per year	WHO [4]
Indian cities exceeding safe air quality limits	Over 70%	CPCB report (2023)
Estimated annual economic loss to India	About $150 billion (≈3% of GDP)	Project study estimate; see also [5]
Share of urban air pollution from vehicles	About 30%	Project study estimate
Share of urban PM2.5 from construction and dust	20–30%	Project study estimate

Table 1.1: Key statistics motivating the project. Estimates differ between studies and methods.

1.7 Organization of the Report

Chapter 2 reviews the literature and existing systems and analyses the problem. Chapter 3 describes the technologies used. Chapter 4 explains the design, implementation and testing of the system. Chapter 5 presents snapshots of the working project. Chapter 6 discusses the future scope, Chapter 7 concludes the report and Chapter 8 lists the references.

1.8 Conclusion

Pollu-Sense addresses the gap between measuring pollution and acting on it. By combining real-time monitoring, AI-based prediction, automated alerts and public participation, it offers a practical foundation for smarter and more accountable air quality management.

Chapter 2

Literature Survey and Problem Analysis

2.1 Introduction

Before designing Pollu-Sense, the causes and effects of air pollution, the limitations of existing monitoring practices and the work done by other researchers were studied. This chapter summarizes that study and defines the problem that the project sets out to solve.

2.2 Major Sources of Air Pollution

- **Industrial Emissions:** Factories release carbon dioxide, sulphur dioxide and oxides of nitrogen into the air.
- **Vehicular Pollution:** Transportation contributes roughly 30% of the total air pollution in cities.
- **Construction and Dust:** Construction activity accounts for about 20–30% of PM2.5 levels in urban areas.
- **Burning of Fossil Fuels:** Coal-fired power plants are a major contributor.
- **Agricultural Activities:** Stubble burning sharply increases pollution, especially in North India.
- **Household Pollution:** Biomass fuels used for cooking in rural areas add to indoor and outdoor pollution.

2.3 Health and Economic Impact

Long-term exposure to fine particulate matter increases the risk of respiratory infections such as asthma, bronchitis, chronic obstructive pulmonary disease (COPD) and lung infections. It contributes to lung cancer because of carcinogenic pollutants, and it is associated with cardiovascular disease, adverse pregnancy outcomes and the worsening of existing health conditions [4].

Delhi, Mumbai, Kolkata and the Indo-Gangetic Plain face severe pollution due to industrial activity, vehicular emissions and agricultural practices such as stubble burning. The economic impact includes higher healthcare costs, reduced workforce productivity and loss of labour income through premature deaths; the World Bank and IHME estimate the welfare losses to be a very large share of national income [5].

2.4 Existing Practices and Their Limitations

Table 2.1 compares the current practice in most cities with the approach proposed in Pollu-Sense.

Aspect	Existing practice	Pollu-Sense approach
Monitoring	Periodic reports from a limited number of fixed stations.	Continuous, real-time tracking using sensors, APIs and satellite-ready data feeds.
Prediction	Mostly reactive; levels are reported after they occur.	Machine-learning models forecast spikes so that preventive measures can be taken.
Source identification	High levels are detected but not attributed to a polluter.	AI analysis points to traffic, industry, construction or natural causes.
Alerts and action	Delayed action and compliance issues.	Automated e-mail alerts with escalation to authorities if there is no response.
Public participation	Limited; citizens are only information consumers.	Crowdsourced reporting, newsletter, reward ideas and health guidance.
Compliance tracking	Manual inspections with little follow-up.	Historical records and a compliance scorecard for each industry.

Table 2.1: Comparison of existing practice and the proposed system.

2.5 Related Work

Sulaimon et al. [1] present a review of machine-learning techniques used for air pollution prediction, and Palanichamy et al. [2] examine machine-learning methods for forecasting PM2.5. Both indicate that data-driven models can forecast pollutant concentrations from historical and meteorological data, which supports the prediction module of Pollu-Sense. Guerrón et al. [3] review IoT sensor nodes for air pollution monitoring, which motivates the sensor-based data collection layer of this project.

Further studies referred to during this project discuss artificial-intelligence-assisted air quality monitoring for smart-city management [15], low-cost IoT-based pollution monitoring systems [16], and the value of crowdsourced pollution data for public health [17]. Researchers at the Johns Hopkins Applied Physics Laboratory are also using AI to simulate atmospheric models so that air quality forecasts become faster and more accurate [18]. Pollu-Sense draws on all of these ideas in a single integrated platform.

2.6 National Air Quality Index (AQI)

In India, air quality is communicated through the National Air Quality Index, which converts the concentrations of several pollutants into a single number and a colour-coded category [8]. Pollu-Sense uses the same idea so that readings are easy for citizens to understand and so that alert thresholds can be tied to AQI categories.

AQI category	AQI range	Possible health impact
Good	0–50	Minimal impact.
Satisfactory	51–100	Minor breathing discomfort to sensitive people.
Moderately polluted	101–200	Breathing discomfort to people with lung or heart disease, children and older adults.
Poor	201–300	Breathing discomfort to most people on prolonged exposure.
Very poor	301–400	Respiratory illness on prolonged exposure.
Severe	401–500	Affects healthy people and seriously affects those with existing diseases.

Table 2.2: National AQI categories (CPCB) [8].

2.7 Legal and Regulatory Framework

The Air (Prevention and Control of Pollution) Act, 1981 gives the Central Pollution Control Board a leading role in preventing and controlling air pollution and assigns responsibilities to the State Pollution Control Boards [6]. The Environment (Protection) Act, 1986 empowers the Government of India to protect and improve environmental quality, to control and reduce pollution from all sources, and to prohibit or restrict the operation of industrial facilities on environmental grounds [7]. Pollu-Sense supports this framework by supplying timely, time-stamped evidence of pollution events and repeat violations to the regulating authorities.

2.8 Conclusion

The study shows that the central gaps in air quality management are the lack of real-time monitoring, the inability to identify polluters and the delay between detection and action. These gaps define the problem that Pollu-Sense is designed to solve.

Chapter 3

Technology Specification

3.1 Introduction

Pollu-Sense was built as a full stack, AI-enabled web application. The technology choices were guided by four needs: real-time data handling, accuracy of prediction, a responsive user interface and low-cost, scalable deployment. The project follows the complete software development life cycle (SDLC): requirement analysis, system design, implementation, testing and deployment.

3.2 Technology Description

The system uses the following technologies, grouped by the layer of the application in which they are used.

Front End

React.js: A component-based JavaScript library used to build the single-page web interface: home page, dashboard, map, about and contact pages.

Tailwind CSS: A utility-first CSS framework used for a clean, consistent and mobile-responsive design.

Chart.js: Used to draw the real-time trend charts for humidity, temperature and AQI on the dashboard [13].

Google Maps API: Used for the interactive map, on which each monitoring location is marked with its AQI and pollutant readings [10].

Back End and Real-Time Services

Django (Python): The main back-end framework, used for the REST APIs, business logic, user management and the data layer [12].

Flask with WebSockets: A lightweight API service that pushes live readings and alerts to the browser the moment new data arrives.

Firebase Authentication and Firebase Cloud Messaging: Provide secure sign-in and instant notification delivery [11].

Artificial Intelligence and Data

TensorFlow: A deep-learning framework used to train models that predict future pollution levels from time-series data [14].

Scikit-learn: A machine-learning library used for data pre-processing and for classical models such as regression and random forests, and for source classification [9].

PostgreSQL: A relational database used to store sensor readings and historical pollution patterns, which also form the basis of the industry compliance scorecard.

OpenWeatherMap API and CPCB data: External sources of current and historical air quality and weather data [10].

IoT sensors (Arduino-based): Gas, particulate, temperature and humidity sensors that supply local, real-time readings [3].

Deployment

Render (free tier): Used to host the application. The Google Maps API is integrated for visualization, and the design allows smart-city systems to be added later.

The combination of these tools allowed the team to build an application that is modular, scalable and secure. Each layer can be replaced or extended independently, for example by adding new sensors or a different prediction model, without redesigning the whole system.

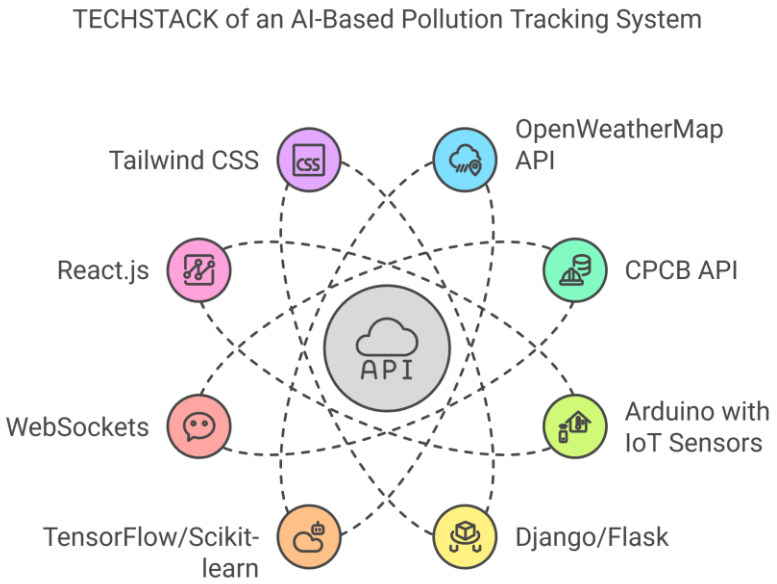


Figure 3.1: Technology stack of the Pollu-Sense system.

3.3 System Requirements

Table 3.1 lists the hardware and software required to develop and run the project.

Category	Specification
Hardware (development)	Computer with at least 8 GB RAM and a multi-core processor; stable internet connection.
Hardware (sensing)	Arduino-compatible microcontroller with particulate, gas, temperature and humidity sensors.
Front end	React.js, Tailwind CSS, Chart.js, Google Maps API.
Back end	Python, Django, Flask, WebSockets, PostgreSQL.
AI / ML	TensorFlow, Scikit-learn, pandas, NumPy.
Cloud and services	Firebase (Authentication, Cloud Messaging), OpenWeatherMap API, Render hosting.
Tools	VS Code, Git and GitHub, a modern web browser.

Table 3.1: Hardware and software requirements.

3.4 Exposure Level

Front-end development: Designing responsive and interactive interfaces with React.js and Tailwind CSS, including dashboards, charts and maps.

Back-end development: Building REST APIs and real-time services with Django, Flask and WebSockets, and managing data in PostgreSQL.

AI and data: Collecting and cleaning data, training and evaluating prediction models with TensorFlow and Scikit-learn.

Cloud and IoT: Using Firebase for authentication and notifications, integrating sensor data and deploying on a cloud platform.

3.5 Conclusion

The selected technologies cover the whole path from sensor to user: data collection, intelligent processing, real-time delivery and visualization. Working with them gave the team practical, end-to-end experience of building a data-driven web application.

Chapter 4

Project Description

4.1 Introduction

The development of Pollu-Sense is the practical application of the concepts of web development, data science and the Internet of Things learned during the course. The system automates air quality tracking and alerting, reduces manual intervention, and provides the public and the authorities with trustworthy, up-to-date information.

The primary goal is to build a scalable and intelligent application that detects pollution spikes in different regions, identifies their likely source, notifies the responsible parties and, if necessary, escalates the case to the regulators.

4.2 Project Description

Pollu-Sense is a web-based platform with three kinds of users:

- **Citizen / Public user:** Views live AQI, temperature and humidity on the dashboard and map, reads health guidance, subscribes to the newsletter and can report pollution sources.
- **Industry / Responsible party:** Receives automated alerts when pollution attributed to it crosses the threshold and is expected to take corrective action.
- **Authority / Administrator:** Receives escalation reports, studies historical patterns and compliance scores, and decides on regulatory action.

Key Features:

- **Real-Time AI Detection:** Uses sensors and open data to detect pollution spikes in different regions.
- **Source Identification:** AI algorithms analyze data to decide whether pollution comes from traffic, industries or natural causes.
- **Automated Notifications:** If pollution exceeds a threshold, the responsible industry or individual is alerted by e-mail.
- **Escalation Mechanism:** If no action is taken, authorities receive a report for regulatory intervention.
- **Predictive Analysis:** Forecasts future pollution levels so that preventive measures can start early.
- **Crowdsourced Reporting:** Citizens contribute real-time pollution information, which increases transparency.
- **Compliance Scorecard:** Assigns environmental ratings to industries based on real-time and historical data.
- **IoT-Powered Response:** Smart air purifiers such as mist sprays and filters can be activated in high-pollution zones.

4.3 System Analysis and Design

The system was analysed and designed using standard software engineering principles.

1. **Requirement Analysis:** The functional requirements include live monitoring, prediction, alerting, escalation, map and dashboard views, newsletter subscription and a contact facility. The non-functional requirements include real-time response, reliability, security of user data, scalability and a responsive interface.
2. **System Design:** The front end is a single-page application built from reusable React components. The back end exposes REST and WebSocket endpoints, and the AI module runs as a separate service that reads from and writes to the database.
3. **Database Design:** PostgreSQL stores locations, sensor readings, predictions, alerts and subscriber details in structured tables so that historical patterns can be queried quickly.
4. **Flow Diagrams and Use Cases:** Flowcharts and use-case diagrams were prepared to visualize data flow, user actions and the escalation path, as shown in Figure 4.1 and Table 4.1.
5. **Security and Validation:** Authentication, input validation and access control were included to protect user data and prevent misuse.

The design is modular and scalable: new cities, sensors, data sources and smart-city integrations can be added without changing the core of the system.

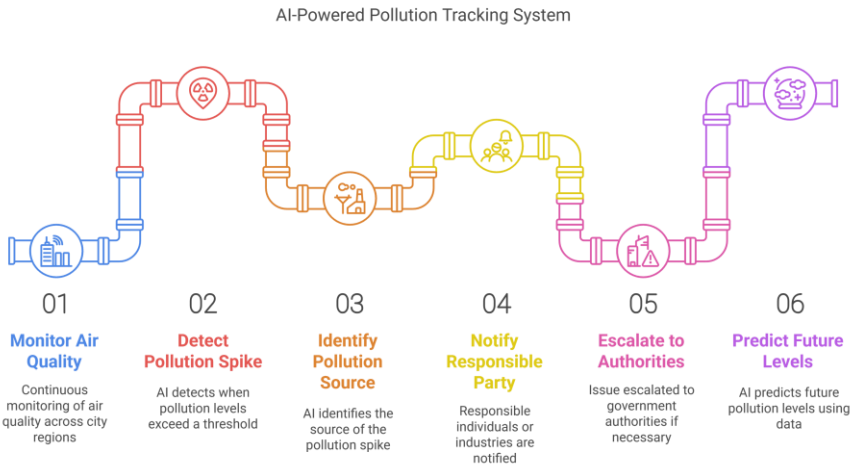


Figure 4.1: Flow chart of the AI-powered pollution tracking system.

Table 4.1: Actors of the system and their main use cases.

Actor	Main use cases
Citizen	View live AQI and map, read health advice, subscribe, report pollution.
Industry	Receive alerts, take corrective action, view own compliance score.
Authority	Receive escalations, view history and hotspots, decide regulatory action.
AI module	Detect spikes, identify sources, predict levels, trigger alerts.

4.4 System Architecture

Pollu-Sense is organised into five layers, each with a clear responsibility.

1. **Sensing layer:** IoT sensors and external APIs (OpenWeatherMap, CPCB) supply raw air quality and weather data.
2. **Data layer:** The data is cleaned, validated and stored in PostgreSQL, together with historical records.
3. **Intelligence layer:** TensorFlow and Scikit-learn models predict pollution levels, detect spikes and classify the probable source.
4. **Application layer:** Django and Flask services expose APIs and WebSockets to the React front end, which shows the dashboard, trends and map.
5. **Notification layer:** Alerts, e-mails, newsletters and escalation reports are generated and delivered.

4.5 Working of the System

The system works as a continuous cycle with the following stages.

- **Stage 1 – Monitor air quality:** Sensors and data feeds provide continuous readings for each region.
- **Stage 2 – Detect pollution spike:** The AI model compares live readings against thresholds and learned patterns, and flags an abnormal rise.
- **Stage 3 – Identify pollution source:** Pattern analysis indicates whether the spike is linked to traffic, an industrial area, construction or a natural cause.
- **Stage 4 – Notify the responsible party:** An automated e-mail alert with the AQI value, time and location is sent to the concerned industry or individual.
- **Stage 5 – Escalate to authorities:** If no corrective action is seen within the defined time, a report is forwarded to the regulating authority.
- **Stage 6 – Predict future pollution levels:** Collected data is fed back to the model to forecast the coming hours and days, which allows preventive steps.

If an industry resumes polluting after it has cleared an inspection, the system detects the repeat violation and alerts the authorities immediately, because the historical record is always available.

4.6 System Development / Implementation

The development phase translated the design into a working system through six modules, following the project roadmap.

Module 1: Data Collection and Sensor Integration

- Data is gathered from the OpenWeatherMap API, CPCB data sets and IoT sensors for real-time pollution tracking.

Module 2: Back-End Development and AI Model Training

- Django (Python) provides the APIs; TensorFlow and Scikit-learn models are trained for prediction and source classification.

Module 3: Front End and User Interaction Platform

- A React.js website with Firebase authentication, a dashboard, an interactive map, newsletter subscription and crowdsourced reporting.

Module 4: Real-Time Monitoring and Alerts System

- A Flask API with WebSockets delivers live updates, and Firebase Cloud Messaging and e-mail send instant notifications.

Module 5: Industry Compliance and Historical Tracking

- Pollution patterns are stored in PostgreSQL and visualized with Chart.js dashboards; each industry receives a compliance rating.

Module 6: Final Deployment and Smart-City Integration

- The application is hosted on Render (free tier) and integrated with the Google Maps API for visualization.

Features Implemented

- Home page with a live AQI summary, ‘How It Works’ and ‘Advanced Features’ sections.
- Interactive map with AQI markers, location search and pollutant-wise readings.
- Dashboard with humidity, temperature and AQI cards and real-time trend charts.
- Automated alert e-mail when AQI crosses the threshold.
- Newsletter subscription with welcome e-mail, and a contact form.

4.7 System Testing / Maintenance

Testing:

The system was tested at several levels to ensure quality, reliability and functionality.

- **Unit Testing:** Individual modules, API endpoints and components were tested separately.
- **Integration Testing:** The interaction between the front end, back end, database and external APIs was verified.
- **System Testing:** The complete workflow, from data collection to alert delivery, was tested end to end.
- **User Acceptance Testing (UAT):** Users checked usability, responsiveness and overall performance.

Table 4.2 lists the main scenarios that were checked.

Test scenario	Expected behaviour
Live data reaches the dashboard	AQI, temperature and humidity cards update without a page reload.
AQI crosses the configured threshold	An alert e-mail with the AQI value is sent to the subscriber.
A user subscribes to the newsletter	A welcome e-mail is received and the subscription is recorded.
A location is searched on the map	The map moves to the location and shows its AQI details.
The site is opened on a phone	The layout adapts to the screen size and stays readable.

Table 4.2: Test scenarios and expected behaviour.

Maintenance:

The modular code makes it easy to fix bugs, add sensors and update the models. Monitoring of API limits and sensor health is part of routine maintenance, and the models can be retrained periodically as new data arrives.

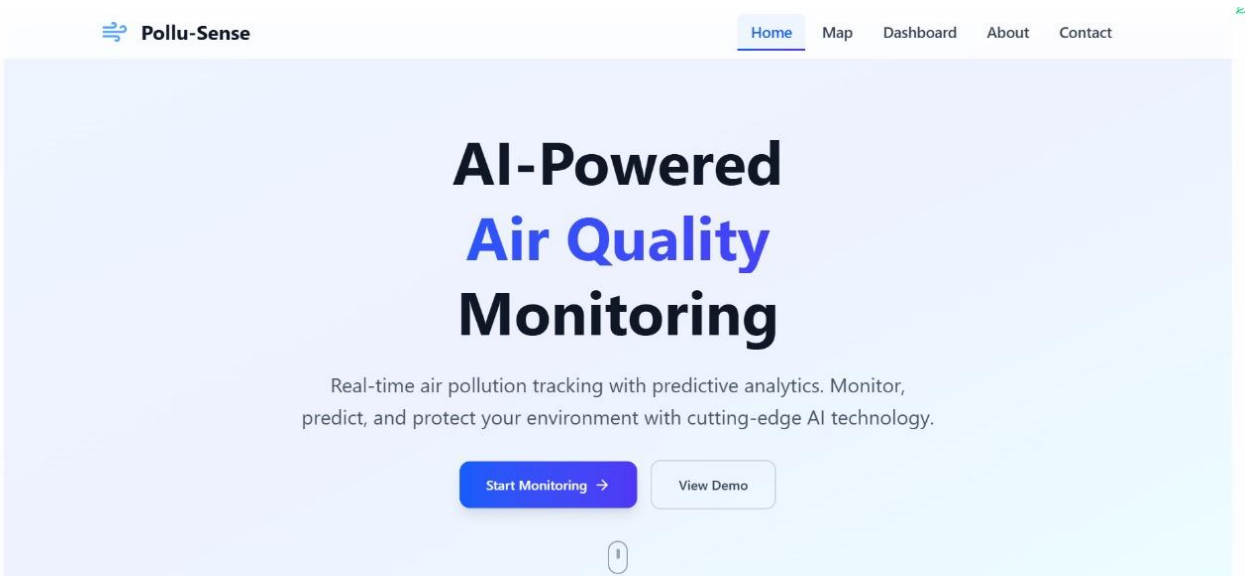
4.8 Conclusion

Pollu-Sense successfully demonstrates the use of AI, IoT and full stack web technologies to monitor air quality, predict pollution and trigger timely action. It reduces manual effort, improves transparency and provides a platform that can be scaled to more cities and pollution sources.

Chapter 5

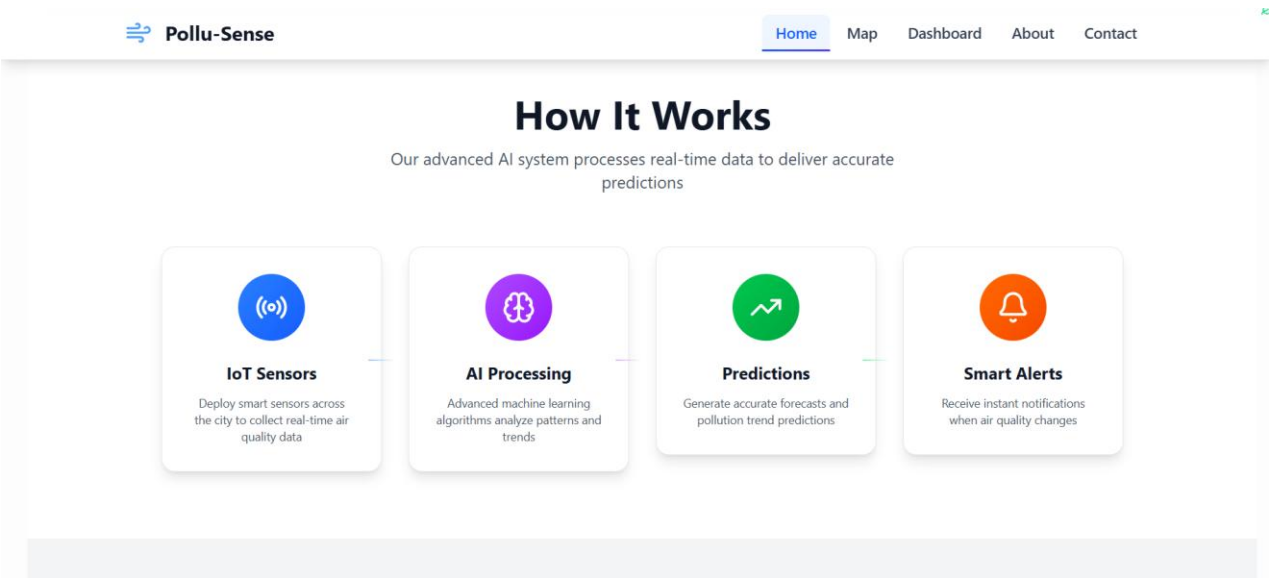
Snapshots of the Project

5.1 Home Page



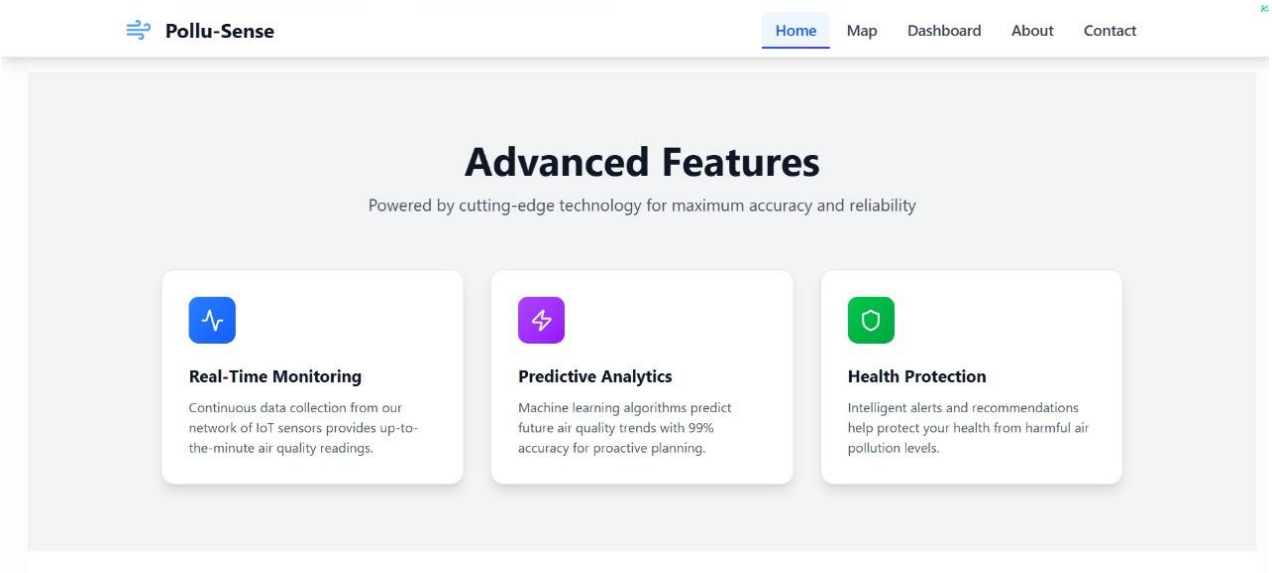
Landing page with the headline, call-to-action buttons and navigation to Map, Dashboard, About and Contact.

5.2 How It Works



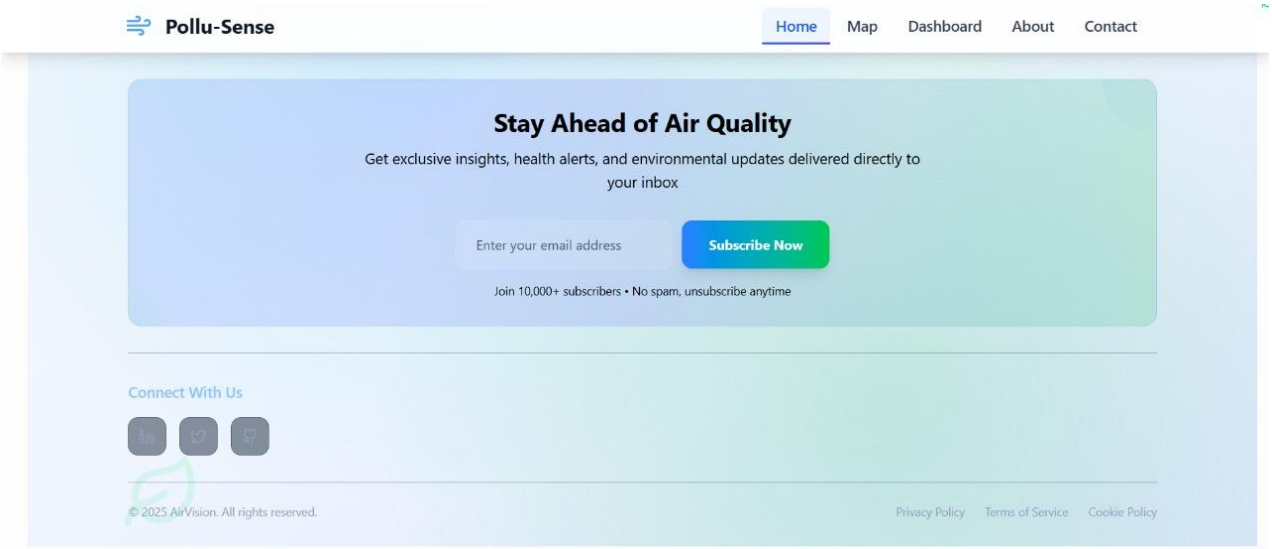
Four-step explanation: IoT sensors, AI processing, predictions and smart alerts.

5.3 Advanced Features



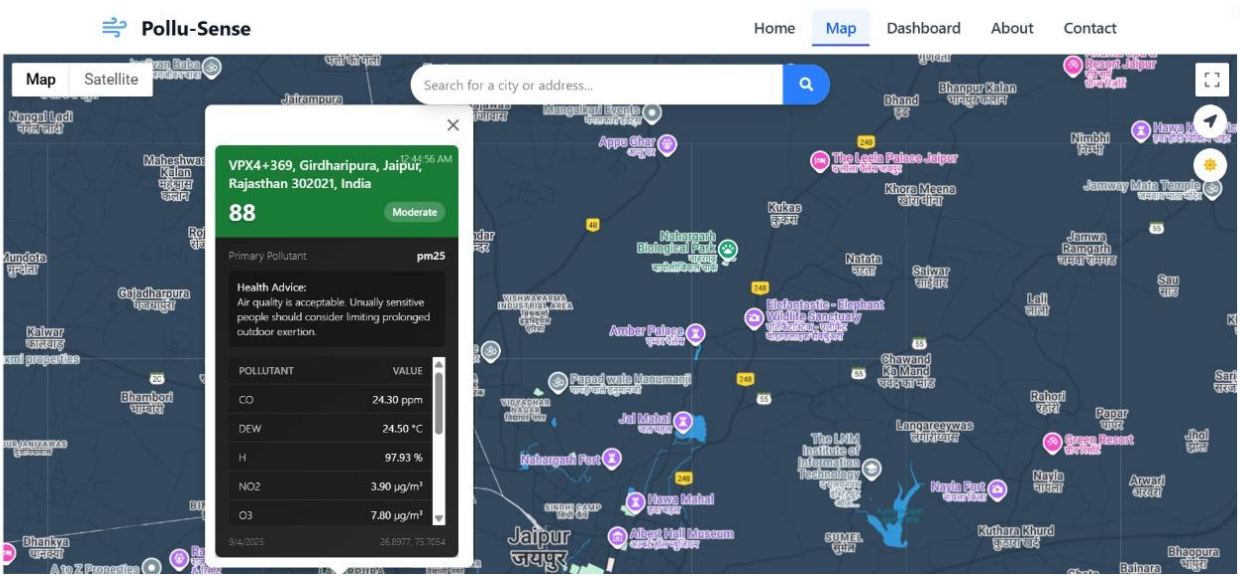
Real-time monitoring, predictive analytics and health protection.

5.4 Newsletter Subscription Section



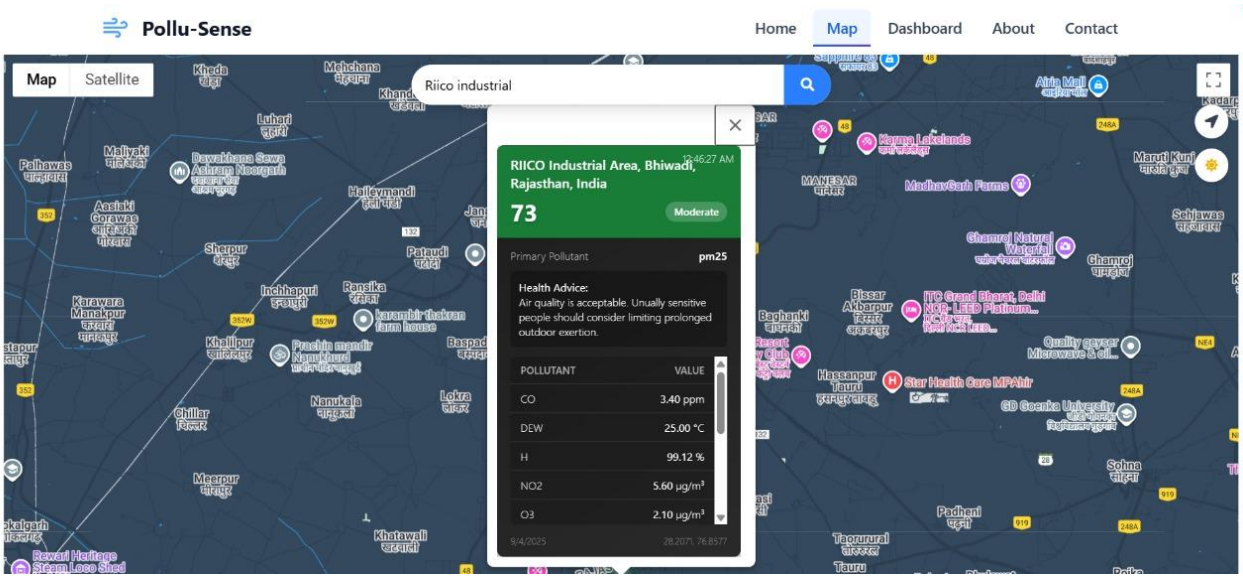
Section that lets visitors subscribe to air quality updates by e-mail.

5.5 Interactive Pollution Map



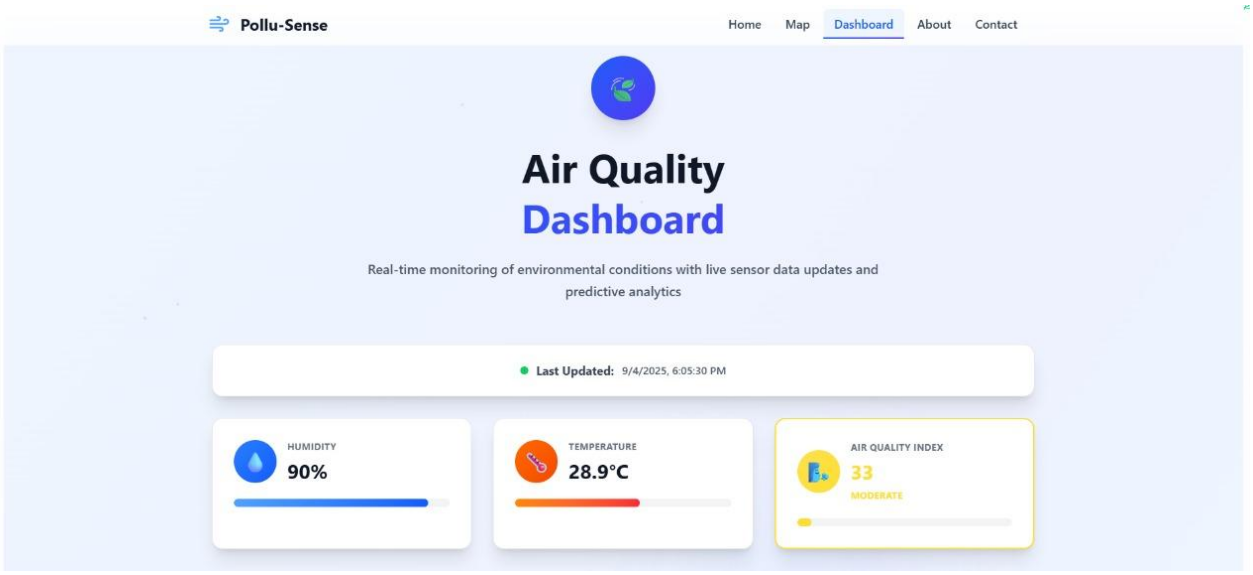
Map view with monitoring locations and a detail card showing AQI and pollutant readings.

5.6 Location Search on the Map



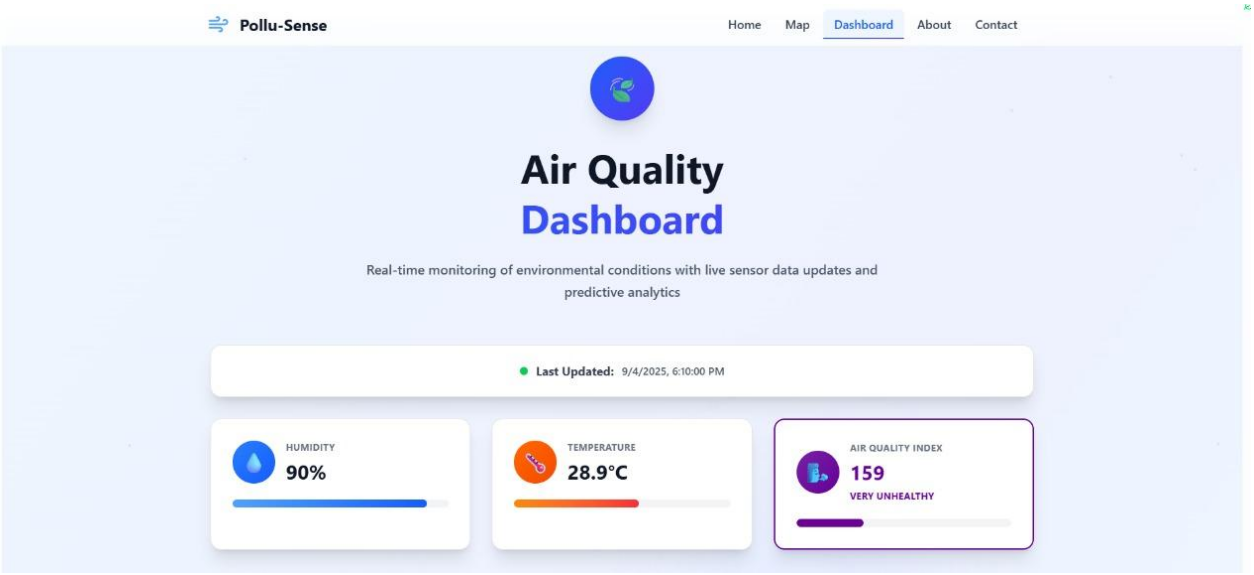
Searching for a location shows its AQI category and the individual pollutant concentrations.

5.7 Air Quality Dashboard



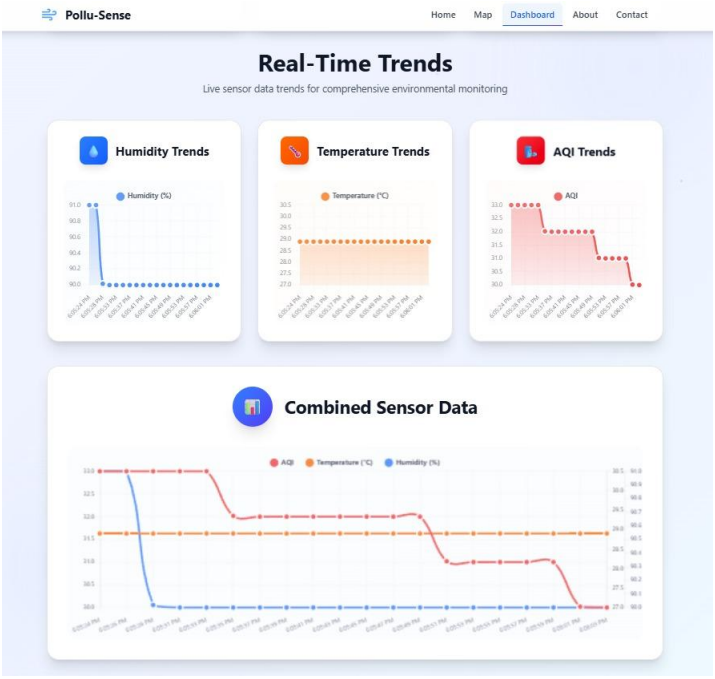
Dashboard cards for humidity, temperature and the air quality index.

5.8 Dashboard at a Higher AQI



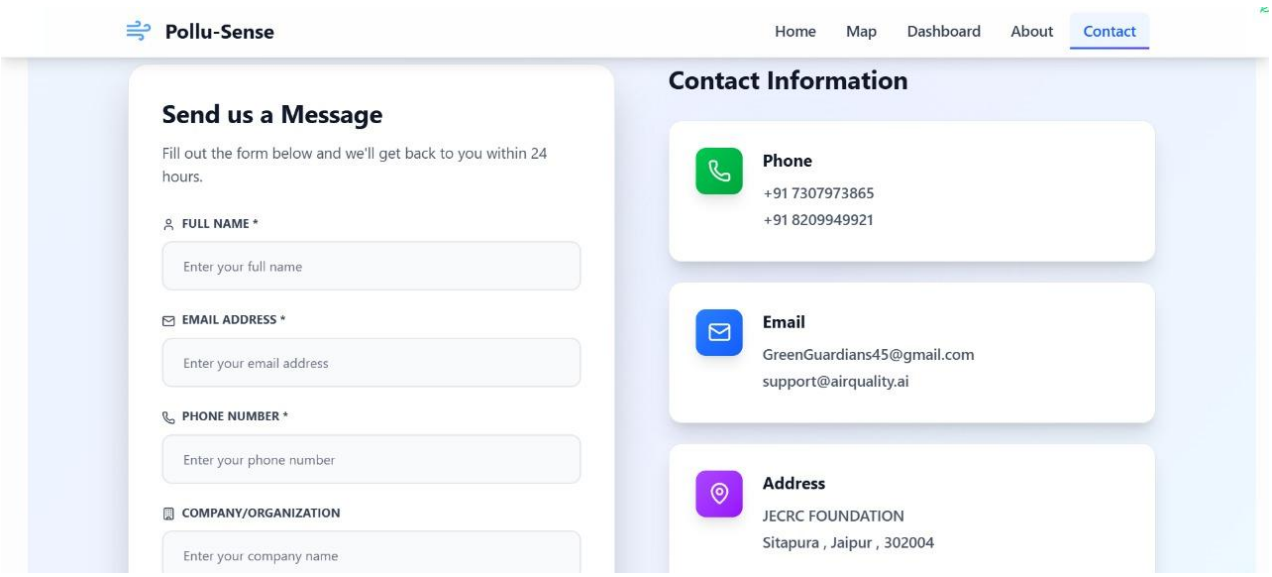
The AQI card changes its status and colour when the index rises.

5.9 Real-Time Trends



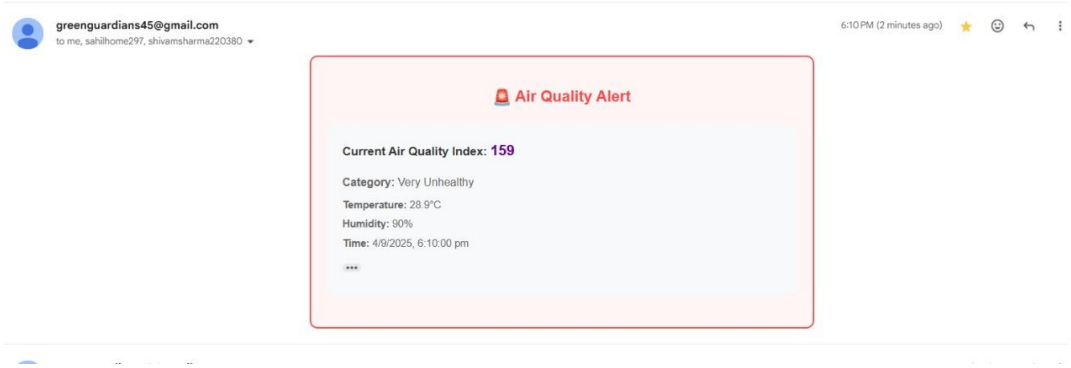
Trend charts for humidity, temperature and AQI, with a combined sensor data graph.

5.10 Contact Page



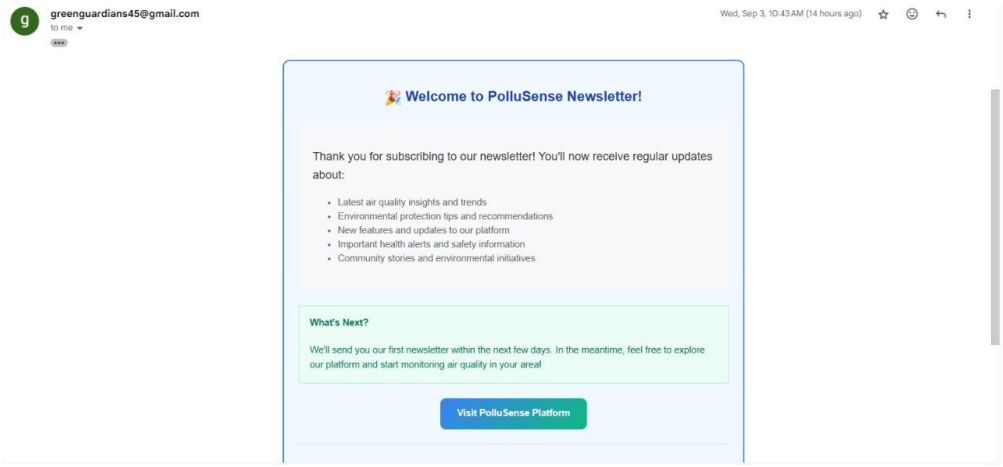
Contact form and contact information for queries and feedback.

5.11 Automated Air Quality Alert E-mail



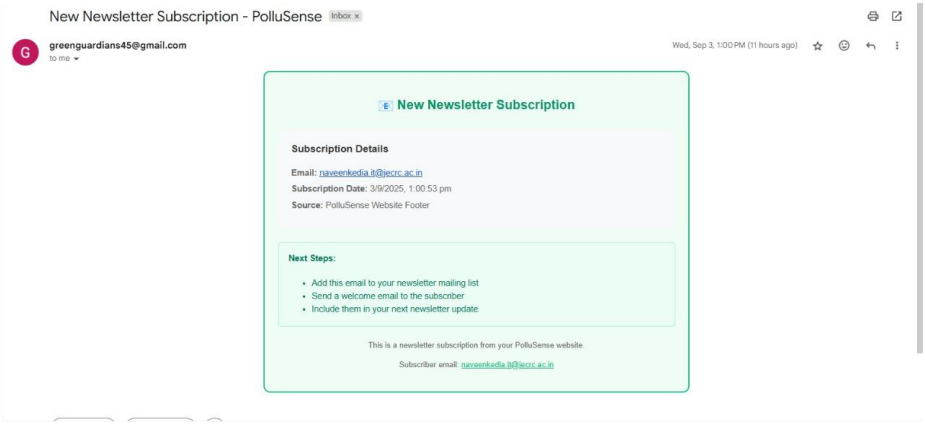
Alert e-mail showing the current AQI, category, temperature, humidity and time.

5.12 Newsletter Welcome E-mail



Welcome message received after subscribing to the Pollu-Sense newsletter.

5.13 New Subscription Notification



Notification to the administrator when a new subscriber joins.

Chapter 6

Future Scope of the Project

Pollu-Sense establishes a working base for AI-driven air quality management. The current prototype already covers monitoring, prediction, alerting and public engagement, but there is wide scope to make it more accurate, more inclusive and more widely adopted. The following enhancements are proposed.

1. Dedicated Mobile Applications

Native Android and iOS apps can deliver push notifications about unhealthy air, location-based advisories and offline access to recent readings. A mobile-first approach would improve reach and make crowdsourced reporting easier by allowing citizens to attach photos and locations of pollution sources.

2. Satellite and Remote-Sensing Data

Combining ground sensors with satellite observations can fill the gaps between monitoring stations and provide city-wide coverage, especially for stubble burning and large-scale industrial emissions.

3. Advanced Deep-Learning Forecasting

Hybrid models such as LSTM-based networks combined with meteorological and traffic data can improve the accuracy of forecasts and provide hyper-local, multi-day predictions, building on current research on AI-based air quality forecasting [1], [2], [18].

4. Blockchain-Based Transparency

Recording alerts, inspection results and compliance scores on a blockchain would make them tamper-proof, which increases the trust of the public and the regulators in the data.

5. Larger and Calibrated Sensor Network

Deploying more low-cost sensor nodes, with regular calibration against reference-grade stations, will increase the spatial resolution of the data and the reliability of the source identification [3].

6. Deeper Smart-City Integration

The platform can be linked with traffic management, urban planning and emergency-response systems so that, for example, traffic can be diverted or construction activity paused when pollution is forecast to reach dangerous levels. Automated mist sprays and air filters in critical zones can be triggered directly by the system.

7. Gamified Citizen Participation and Personal Health Advice

A reward-based system can encourage citizens to report pollution sources and to use public transport. A personal pollution-footprint calculator can track individual exposure and suggest healthier choices, including route and time advice for people with respiratory conditions.

8. Multi-Language and Accessibility Support

Support for Hindi and other regional languages, voice navigation, high-contrast themes and screen-reader compatibility would make the system usable for a much wider population, including differently-abled users.

9. Regulator Analytics and Policy Support

An advanced analytics dashboard for authorities could provide hotspot maps, repeat-offender lists, seasonal trends and the estimated effect of policies, helping them to plan interventions and to measure results.

10. Expansion to Multiple Cities and Sources

The same architecture can be extended from the first pilot city to many cities and adapted to different pollution sources, such as industrial, vehicular, construction or agricultural emissions.

Chapter 7

Conclusion

The Pollu-Sense project, developed as the Minor Project of the Department of Information Technology, Poornima College of Engineering, is the practical implementation of the knowledge acquired in web development, machine learning and the Internet of Things. The project creates an intelligent, transparent and scalable platform for monitoring air quality, predicting pollution and prompting timely corrective action.

The system integrates React.js, Tailwind CSS, Django, Flask, TensorFlow, Scikit-learn, PostgreSQL, Firebase, OpenWeatherMap, CPCB data and the Google Maps API into a single working application. Through this project the team gained hands-on experience in real-time data handling, AI-based prediction, alert and escalation design, and cloud deployment.

This chapter summarizes the lessons learned, the skills developed, the limitations identified and the recommendations for future enhancement.

7.1 Lessons Learned and Skills Developed

Technical Skills

- **Front-end development:** Responsive interfaces, reusable components, charts and map integration using React.js, Tailwind CSS, Chart.js and Google Maps.
- **Back-end development:** REST APIs, WebSockets and server-side logic using Django and Flask.
- **Machine learning:** Preparing data, training and evaluating prediction models with TensorFlow and Scikit-learn.
- **Database management:** Designing PostgreSQL schemas for time-series sensor data and historical analysis.
- **Cloud and integration:** Using Firebase, third-party APIs and free-tier hosting to deploy a working product.
- **Debugging and optimization:** Finding and fixing issues across the whole stack, from sensor input to the browser.

Communication Skills

- Documenting requirements, workflows and code clearly, and presenting the project with a demonstration and a video.
- Explaining technical issues, solutions and progress to faculty coordinators and peers.

Interpersonal and Professional Skills

- **Team collaboration:** Dividing modules among four members, reviewing each other's work and integrating the result.
- **Time management:** Planning tasks and meeting deadlines while managing several modules at the same time.
- **Critical thinking:** Analysing requirements, identifying risks and proposing practical solutions.

Overall, the project offered a complete learning experience that combined technical, communication and professional skills.

7.2 Limitations and Recommendations

Limitations

- The accuracy of predictions and source identification depends on the amount, quality and coverage of the available sensor and historical data.
- Source attribution is probabilistic and should be confirmed by field inspection before any penalty is imposed.
- The system depends on third-party APIs and free-tier hosting, which have usage limits and may affect availability.
- The current version is a web application only; a mobile app and offline support are not yet available.
- Support for regional languages and accessibility features is limited.

Recommendations

- **Wider and calibrated sensor network:** To improve coverage and reliability of readings.
- **Advanced models:** To use deep-learning and hybrid approaches with more historical data.
- **Blockchain and audit trail:** To make compliance records tamper-proof.
- **Mobile app and multilingual interface:** To reach more citizens.
- **Scalability optimization:** To strengthen the back end for large-scale, multi-city deployment.

By implementing these upgrades, Pollu-Sense can evolve into a comprehensive smart-city platform that supports regulators, industries and citizens in the shared goal of cleaner air.

Chapter 8

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Nirmal Todwal

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